Python Course Syllabus

Description: Python curriculum from fundamentals through data structures, functions, object-oriented programming, databases, APIs, and project development.

Module 1: Python Fundamentals

Module Description: Introduction to Python, its features, installation, syntax, and essential programming concepts.

Topic 1: Introduction to Python

Topic Description:

Lesson 1: What is Python?

Lesson Description: Understand Python and its common applications.

Lesson 2: Python Features

Lesson Description: Explore readability, portability, and the Python ecosystem.

Lesson 3: Installing Python and an IDE

Lesson Description: Install Python and configure a code editor.

Topic 2: Python Basics

Topic Description:

Lesson 1: Running Python Programs

Lesson Description: Run scripts and use the interactive interpreter.

Lesson 2: Variables and Data Types

Lesson Description: Work with numbers, strings, booleans, and type conversion.

Lesson 3: Operators and Input/Output

Lesson Description: Use operators, input(), and print().

Module 2: Control Flow and Data Structures

Module Description: Write conditional logic and loops and organize data using Python collections.

Topic 1: Control Flow

Topic Description:

Lesson 1: Conditional Statements

Lesson Description: Use if, elif, and else to control execution.

Lesson 2: For and While Loops

Lesson Description: Repeat tasks with for and while loops.

Lesson 3: Break, Continue, and Range

Lesson Description: Control loop behavior and generate sequences.

Topic 2: Python Collections

Topic Description:

Lesson 1: Lists and Tuples

Lesson Description: Store and access ordered collections.

Lesson 2: Sets

Lesson Description: Work with unique values and set operations.

Lesson 3: Dictionaries

Lesson Description: Manage key-value pairs.

Lesson 4: Comprehensions

Lesson Description: Build collections with concise expressions.

Module 3: Functions, Modules and File Handling

Module Description: Create reusable functions, organize Python programs, and handle files and errors.

Topic 1: Functions and Modules

Topic Description:

Lesson 1: Defining Functions

Lesson Description: Create functions with parameters and return values.

Lesson 2: Arguments and Scope

Lesson Description: Understand positional, keyword, default arguments, and variable scope.

Lesson 3: Modules and Packages

Lesson Description: Import standard-library and custom modules.

Topic 2: Files and Exceptions

Topic Description:

Lesson 1: Reading Files

Lesson Description: Open and read text files safely.

Lesson 2: Writing Files

Lesson Description: Create and update files using context managers.

Lesson 3: Exception Handling

Lesson Description: Handle errors with try, except, else, and finally.

Lesson 4: Working with JSON

Lesson Description: Read and write JSON data.

Module 4: Object-Oriented Programming in Python

Module Description: Model applications with classes, objects, and core object-oriented concepts.

Topic 1: Classes and Objects

Topic Description:

Lesson 1: Creating Classes

Lesson Description: Define classes and instantiate objects.

Lesson 2: Constructors and Attributes

Lesson Description: Initialize object state with __init__.

Lesson 3: Instance and Class Methods

Lesson Description: Use instance, class, and static methods.

Topic 2: OOP Principles

Topic Description:

Lesson 1: Inheritance

Lesson Description: Reuse and extend behavior through inheritance.

Lesson 2: Polymorphism

Lesson Description: Implement shared interfaces with different behavior.

Lesson 3: Encapsulation

Lesson Description: Organize and protect object state.

Lesson 4: Special Methods

Lesson Description: Use Python dunder methods for custom object behavior.

Module 5: Advanced Python, Databases and APIs

Module Description: Apply advanced Python features and build small applications using databases and web APIs.

Topic 1: Advanced Python Concepts

Topic Description:

Lesson 1: Iterators and Generators

Lesson Description: Process sequences lazily using iterators and yield.

Lesson 2: Decorators

Lesson Description: Wrap functions to extend their behavior.

Lesson 3: Virtual Environments and pip

Lesson Description: Manage project environments and dependencies.

Topic 2: Database and API Integration

Topic Description:

Lesson 1: Connecting to SQLite

Lesson Description: Connect Python applications to a relational database.

Lesson 2: Database CRUD Operations

Lesson Description: Create, read, update, and delete records with SQL.

Lesson 3: Working with REST APIs

Lesson Description: Send HTTP requests and process JSON responses.

Topic 3: Project and Deployment Basics

Topic Description:

Lesson 1: Building a Python Project

Lesson Description: Organize a small Python application into reusable files.

Lesson 2: Testing and Logging

Lesson Description: Write basic tests and record application events.

Lesson 3: Deployment Basics

Lesson Description: Prepare dependencies and configuration for running an application.